

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Python** backend applications using **FastAPI** framework with **PostgreSQL** databases, delivering 12+ REST API endpoints for treasury and collateral operations systems supporting 200+ concurrent users with **99.8% uptime SLA**.
- Built full-stack applications combining **Python FastAPI** backend with **React 18** and **TypeScript** frontend using **Material UI** component library, implementing test-driven development practices and reducing feature delivery cycle by **40%**.
- Architected and deployed **AWS Cloud**-based solutions leveraging **EC2, S3, Lambda, and CloudWatch**, integrating microservices via **RESTful Web Services** and service-oriented architecture patterns to enable scalable middle-office operations platforms.
- Designed and conducted comprehensive test strategies using **test-driven development** methodologies, achieving **92% code coverage** across Python modules and detecting critical issues before production release through automated testing pipelines.
- Developed systems documentation and knowledge-base materials supporting third-level support functions, reducing average incident resolution time by **35%** and enabling self-service for **150+ finance operations users** across treasury and trade functions.
- Evaluated and integrated **code generation tools** including **Amazon Q and GitHub Copilot** into development workflow, improving developer productivity by **30%** and contributing to architectural reviews for continuous technology assessment and modernization.

SKILLS

Backend Development – Python, FastAPI, REST APIs, service-oriented architecture, Pydantic, JWT authentication.

Frontend Development – React 18, TypeScript, Material UI, responsive design, component libraries.

Database & Data – PostgreSQL, SQL query optimization, data modelling, ETL pipelines.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, CI/CD pipelines, Jenkins, Git.

Testing & Quality – Test-driven development, automated testing, debugging, code reviews.

Development Tools & AI – Amazon Q, GitHub Copilot, Kiro, AG Grid, AG Charts, Rsbuild, Module Federation.

Domain & Soft Skills – Financial services, treasury operations, collateral management, trade operations, technical documentation, problem-solving, cross-functional collaboration.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Treasury Collateral Operations Platform – Full-Stack Microservices Rebuild (HCL Technologies)

- Led full-stack redesign of legacy treasury and collateral management system, migrating monolithic architecture to **Python FastAPI microservices** deployed on **AWS** with **PostgreSQL** for persistent state and **S3** for document storage, enabling independent scaling of treasury, collateral, and trade operation modules.
- Implemented comprehensive **React 18 + TypeScript** frontend using **Material UI** component library and **AG Grid** for advanced data grids, delivering real-time collateral position dashboards and trade reconciliation views with **sub-500ms** API response times.
- Designed **test-driven development** framework covering **unit, integration, and end-to-end** testing across all microservices, achieving **94% code coverage** and reducing production incidents by **65%** over 9-month delivery cycle.
- Established **CI/CD automation** using **Jenkins pipelines, Git, and Docker** for containerized deployments across development, UAT, and production environments, reducing release cycle from **3 weeks to 4 days** and enabling twice-weekly feature releases.
- Integrated **Amazon Q code generation** tooling into development workflow, accelerating API endpoint generation and data validation layer construction by **28%**, while documenting architectural patterns and runbooks for support teams handling third-level incident escalations.

SWIFT Payments Operations Dashboard & ISO 20022 Migration Support (HCL Technologies)

- Developed production support and real-time monitoring systems for high-volume **SWIFT messaging** platform processing **50,000+ daily payment transactions** across SEPA, cross-border, and instant payment corridors, using **Splunk and CloudWatch** for SLA-critical alerting and transaction-flow visibility.
- Built **Python automation scripts** for payment reconciliation and ISO 20022 message transformation workflows, reducing manual reconciliation overhead by **82%** and ensuring compliance with financial services audit and regulatory requirements including GDPR data protection protocols.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)